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Implementation of a high-order compact finite-difference lattice Boltzmann method in generalized curvilinear coordinates

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ABSTRACT

In this work, the implementation of a high-order compact finite-difference lattice Boltzmann method (CFDLBM) is performed in the generalized curvilinear coordinates to improve the computational efficiency of the solution algorithm to handle curved geometries with non-uniform grids. The incompressible form of the discrete Boltzmann equation with the Bhatnagar–Gross–Krook (BGK) approximation with the pressure as the independent dynamic variable is transformed into the generalized curvilinear coordinates. Herein, the spatial derivatives in the resulting lattice Boltzmann (LB) equation in the computational plane are discretized by using the fourth-order compact finite-difference scheme and the temporal term is discretized with the fourth-order Runge–Kutta scheme to provide an accurate and efficient incompressible flow solver. A high-order spectral-type low-pass compact filter is used to regularize the numerical solution and remove spurious waves generated by boundary conditions, flow non-linearities and grid non-uniformity. All boundary conditions are implemented based on the solution of governing equations in the generalized curvilinear coordinates. The accuracy and efficiency of the solution methodology presented are demonstrated by computing different benchmark steady and unsteady incompressible flow problems. A sensitivity study is also conducted to evaluate the effects of grid size and filtering on the accuracy and convergence rate of the solution. Four test cases considered herein for validating the present computations and demonstrating the accuracy and robustness of the solution algorithm are: unsteady Couette flow and steady flow in a 2-D cavity with non-uniform grid and steady and unsteady flows over a circular cylinder and the NACA0012 hydrofoil at different flow conditions. Results obtained for the above test cases are in good agreement with the existing numerical and experimental results. The study shows the present solution methodology based on the implementation of the high-order compact finite-difference Lattice Boltzmann method (CFDLBM) in the generalized curvilinear coordinates is robust, efficient and accurate for solving steady and unsteady incompressible flows over practical geometries.

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1. Introduction

In the two last decades, there has been much progress in developing the lattice Boltzmann method (LBM) as a powerful technique for computing fluid flow problems. The LBM is based on the kinetic theory in which it studies the particle interactions by using the particle mass distribution function. This mechanism is parallel in nature due to the locality of the dynamic of particles. From a computational point of view, it is well suitable for massively parallel computing. Simplicity of programming and ease of considering microscopic interactions for modeling of additional physical phenomenon are the other advantages of the LBM. The standard LBM with the BGK approximation is restricted to use uniform Cartesian grids with equal spacing. Inherent instability at high Reynolds number flows is also another limitation of this methodology. These two aspects in the standard LBM greatly limit its applications to solve engineering practical problems with complex geometries.

A drawback of using uniform grids is that it needs a high grid resolution in the high gradient regions, such as the thin boundary layer near a solid wall in a high Reynolds number flow. Thus, in order to solve high Reynolds number flows with a sufficient accuracy, a large number of uniform lattices should be used which requires larger computer resources [1]. The study of the treatment of curved or irregular boundaries and the control of mesh around complex geometries are also difficult with uniform Cartesian grids. In Navier–Stokes equation-based solvers, this issue is easily resolved by using non-uniform grids with mesh stretching techniques, often in a curvilinear coordinates system.

The difficulties and instabilities of the conventional LBM from the use of uniform grids are closely related to the strong coupling of the time and space discretizations in the conventional LBM. He and Luo [2] have shown that the discretization of physical space does not necessarily require for coupling with the discretization of momentum space. Based on this idea, alternative LB methods are proposed based on non-uniform grids to overcome the limitations of the conventional LBM to increase the numerical efficiency and accuracy of this methodology as a practical computational fluid dynamics tool. In this way, several extensive efforts have been made to provide improved versions of the LBM which can be classified into two different categories [3]: namely, interpolation types and differential types. In the interpolation strategy, the calculation is based on a coarse grid covering the whole integration domain. Then, a grid refinement technique is used to locally refine the grid in the critical regions where a high grid resolution is needed [3,4]. Another approach is to use a coupling of grids of varying resolution (non-uniform grids) by a local grid interface based on local interpolations [5–7].

In the differential types, the LBM is considered as a particular discretization of the discrete Boltzmann equation (DBE) which is transformed by using the Taylor series expansion in time and space [8]. Some efforts have been made in literature to apply the traditional numerical methods developed in computational fluid dynamics (CFD) for solving the DBE. Finite-difference (FD) [9–18], finite-volume (FV) [19–24], finite-element (FE) [25–27], and recently spectral-element discontinuous Galerkin (SEDG) [28–30] methods have been applied in order to improve computational accuracy and efficiency of the LBM. The differential types LBM methods can overcome the difficulties encountered at high Reynolds numbers by using body-conform, curvilinear meshes with clustering of grid points in critical zones. Although the differential type LBM methods provide such advantages, the locality of the collision of particles is sacrificed when applying these methods because of the stencil in discretizing the spatial derivative terms and the necessity of especial treatments near each boundary. Consequently, the compatibility with parallel computation, that is the main benefit of the conventional LBM, is removed when using the differential type LBM methods.

Among of existing numerical methods to solve the equations governed fluid flows, finite-difference methods have simplicity of implementation and achieving high-order accuracy is rather simple with them compared to other numerical approaches. Handling curved boundaries with these methods is also straightforward and can be easily made by applying a coordinate transformation. The finite-difference LBM is easier to implement and much more efficient to solve than the finite-difference solution of the Navier–Stokes equations. Finite-difference LBM has been extended to the curvilinear coordinates by Mei and Shyy [10]. Guo and Zhao [12] have presented an improved version of the finite-difference LBM proposed by Mei and Shyy with employing a simple explicit scheme for the curvilinear coordinates.

Most of the previous FDLBM developed in literature are based on traditional finite-difference schemes and they may have the stability problems for the calculation of high Reynolds number flows. An accurate and practical numerical method based on the finite-difference LBM is therefore interested that does not contain these deficiencies. In FDLBM, one can apply high-order accurate numerical methods for improving the accuracy and performance of the solution. Recently, we have proposed and applied a high-order compact finite-difference lattice Boltzmann method to solve incompressible flows in Cartesian coordinates with uniform grids [31]. Compared with a great numerical algorithm, compact finite-difference methods are indicated to be significantly more accurate with advantage of having good resolution characteristics and smaller computational stencil size needed. The main objective of this paper is to implement the high-order compact finite-difference scheme to the LBM in the generalized curvilinear coordinates with non-uniform grids. As a result, the geometry of the curved boundaries can be more easily modeled and the solution of flowfield is obtained with high-order accuracy. Note that, similar to other finite-difference LBM solvers, the locality of the collision process is sacrificed due to the stencil in the compact finite-difference scheme applied and the especial treatments needed for implementing boundary conditions. Herein, the spatial derivatives in the incompressible form of LB equation in the computational plane are discretized by using the fourth-order compact finite-difference scheme and the temporal term is discretized by the fourth-order Runge–Kutta scheme to provide an accurate and efficient incompressible flow solver. A high-order spectral-type low-pass compact filter is used to ensure the numerical stability. A sensitivity study is also performed to examine the effects of the numerical parameters

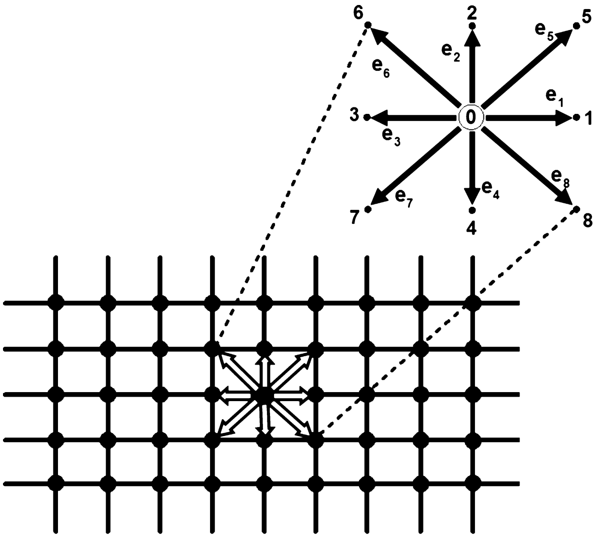

Fig. 1. The D2Q9 lattice and the microscopic velocities.

on the accuracy and performance of the solution. The accuracy and robustness of the high-order compact finite-difference lattice Boltzmann method implemented in the generalized curvilinear coordinates are investigated by solving incompressible flows for different incompressible flow problems at different conditions.

The paper is organized as follows: In Section 2, the incompressible form of the LB equation based on the pressure distribution function as the independent variable is presented and it is transformed into the generalized curvilinear coordinates in Section 3. The high-order compact finite-difference discretization and the Runge–Kutta time stepping scheme for the LB equation in the computational plane are given in Section 4. The filtering scheme used is presented in Section 5 and the boundary conditions required for the numerical scheme for both the macroscopic and microscopic variables are described in Section 6. Finally, Section 7 is devoted to present the numerical results for four incompressible flow problems to examine the accuracy and performance of the solution of the compact finite-difference LBM implemented in the generalized curvilinear coordinates.

2. Governing equations

The main goal of this paper is to implement the fourth-order compact finite-difference lattice Boltzmann scheme in the generalized curvilinear coordinates with non-uniform grids. In the present work, the solution of the incompressible form of the LB equation is obtained on the curved boundaries to assess the accuracy and robustness of the scheme implemented in the body-fitted coordinates. The LB equation governed the particle distribution function $f(t, c, x)$ is used with a single relaxation time and the collision term in the Bhatnagar–Gross–Krook (BGK) approximation [32]

$$\frac{\partial f}{\partial t} + \boldsymbol{e} \cdot \nabla f = -\frac{1}{\tau}\left(f - f^{eq}\right) \tag{1}$$

where τ is the dimensionless collision relaxation time, $\boldsymbol{e}$ is the particle velocity and f^{eq} is the equilibrium distribution function. A two-dimensional square lattice model with nine velocity directions (D2Q9) is employed to discretize Eq. (1) in the lattice configuration. Fig. 1 indicates a rectangular solution domain for the LB equation with the particle distribution function f_α in the direction of the microscopic velocity $\boldsymbol{e}_\alpha$ on Cartesian coordinates

$$\frac{\partial f_\alpha}{\partial t} + \boldsymbol{e}_\alpha \cdot \nabla f_\alpha = -\frac{1}{\tau}\left(f_\alpha - f_\alpha^{eq}\right), \quad \alpha = 0, 1, \ldots, 8 \tag{2}$$

where the subscript α denotes the direction of the particle speed. In the D2Q9 discrete Boltzmann model, the microscopic velocities are given as

$$\boldsymbol{e}_\alpha = (e_{\alpha x}, e_{\alpha y}) = \begin{cases} (0,0) & \alpha = 0 \\ (\cos(\frac{\alpha-1}{2}\pi), \sin(\frac{\alpha-1}{2}\pi))c & \alpha = 1,2,3,4 \\ (\cos(\frac{\alpha-5}{2}\pi + \frac{1}{4}\pi), \sin(\frac{\alpha-5}{2}\pi + \frac{1}{4}\pi))\sqrt{2}c & \alpha = 5,6,7,8 \end{cases} \tag{3}$$

where $c = \Delta x/\Delta t$ is the propagation speed of particles moving between a lattice node and its nearest neighbors and Δx and Δt are the lattice length unit and the time step size, respectively.

The equilibrium distribution function f^{eq} is chosen to satisfy the incompressible Navier–Stokes equations through a Chapman–Enskog expansion procedure. The incompressible form of the LB equation used herein is based on the model given by He and Luo [33]. They introduced a local pressure distribution function and redefining the density distribution in the LB equation by the fact that the error terms in the order of $O(M^2)$ (M is the Mach number) are explicitly removed from the equilibrium distribution function. Therefore, the fluid density is not calculated in the simulations and the incompressible form of the LB equation solves the pressure in the computational domain as the independent variable. In this formulation, the equation of the equilibrium distribution function f_α^{eq} is defined as [33]

$$f_\alpha^{eq} = w_\alpha\left(p + p_0\left(3\frac{\boldsymbol{e}_\alpha \cdot \boldsymbol{u}}{c^2} + \frac{9}{2}\frac{(\boldsymbol{e}_\alpha \cdot \boldsymbol{u})^2}{c^4} - \frac{3}{2}\frac{|\boldsymbol{u}|^2}{c^2}\right)\right) \tag{4}$$

where $\boldsymbol{u} = (u, v)$ is the velocity vector in Cartesian coordinates and the weight coefficient w_α for the D2Q9 model is given by

$$w_\alpha = \begin{cases} \frac{4}{9} & \alpha = 0 \\ \frac{1}{9} & \alpha = 1,2,3,4 \\ \frac{1}{36} & \alpha = 5,6,7,8 \end{cases} \tag{5}$$

The macroscopic fluid pressure p and the macroscopic velocity $\boldsymbol{u}$ are obtained from the following relations

$$p = \sum_\alpha f_\alpha, \qquad p_0\boldsymbol{u} = \sum_\alpha \boldsymbol{e}_\alpha f_\alpha \tag{6}$$

where $p_0 = c_s^2\rho_0$ and ρ_0 is the constant density of the fluid. The incompressible Navier–Stokes equations can be derived from the incompressible form of the LB model through the Chapman–Enskog expansion procedure [33]

$$\frac{1}{c_s^2}\frac{\partial P}{\partial t} + \nabla \cdot \boldsymbol{u} = 0 \tag{7}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla P + \nu \nabla^2 \boldsymbol{u} \tag{8}$$

where $P = \frac{p}{\rho_0}$ is the normalized pressure and ν is the physical value of the kinematic viscosity of the fluid. The relaxation time τ for the finite-difference LB model is defined by the following relation [13]

$$\tau = \frac{\nu}{c_s^2} \tag{9}$$

where $c_s = c\sqrt{\chi}$ is the sound speed and χ is a characteristic of the LB model with a constant value. For the D2Q9 model, $\chi = 1/3$ [34]. This expression of the kinematic viscosity and the relaxation parameter relation indicates that the behavior of the fluid flow does not depend on the discretization of the velocity space in the finite-difference based LB methods.

3. Transformation to generalized curvilinear coordinates

Now, Eq. (2) is transformed into the generalized curvilinear coordinates in which the physical and computational planes are denoted by (x, y) and (ξ, η), respectively:

$$\begin{aligned} \xi &= \xi(x, y) \\ \eta &= \eta(x, y) \end{aligned} \tag{10}$$

The relationship between the physical and computational domains satisfies the following condition

$$\begin{bmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{bmatrix} = \frac{1}{J}\begin{bmatrix} y_\eta & -x_\eta \\ -y_\xi & x_\xi \end{bmatrix} \tag{11}$$

where the Jacobian of the transformation J is denoted by

$$J = x_\xi y_\eta - x_\eta y_\xi \tag{12}$$

Using the chain rule, the convection term in Eq. (2) can be rewritten as

$$\begin{aligned} \boldsymbol{e}_\alpha \cdot \nabla f &= e_{\alpha x}\frac{\partial f_\alpha}{\partial x} + e_{\alpha y}\frac{\partial f_\alpha}{\partial y} = e_{\alpha x}\left(\frac{\partial f_\alpha}{\partial \xi}\xi_x + \frac{\partial f_\alpha}{\partial \eta}\eta_x\right) + e_{\alpha y}\left(\frac{\partial f_\alpha}{\partial \xi}\xi_y + \frac{\partial f_\alpha}{\partial \eta}\eta_y\right) \\ &= (e_{\alpha x}\xi_x + e_{\alpha y}\xi_y)\frac{\partial f_\alpha}{\partial \xi} + (e_{\alpha x}\eta_x + e_{\alpha y}\eta_y)\frac{\partial f_\alpha}{\partial \eta} \\ &= \tilde{e}_{\alpha\xi}\frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha\eta}\frac{\partial f_\alpha}{\partial \eta} \end{aligned} \tag{13}$$

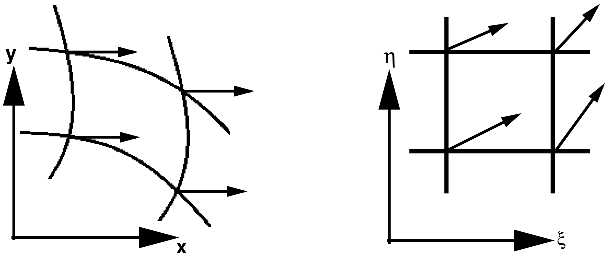

Fig. 2. Microscopic velocity vectors $\boldsymbol{e}_1 = (1, 0)$ in the physical plane (left) and the contravariant velocity vectors $\tilde{\boldsymbol{e}}_1$ in the computational plane (right).

where $\tilde{\boldsymbol{e}}_\alpha = (\tilde{e}_{\alpha\xi}, \tilde{e}_{\alpha\eta}) = (e_{\alpha x}\xi_x + e_{\alpha y}\xi_y, e_{\alpha x}\eta_x + e_{\alpha y}\eta_y)$ is the microscopic contravariant velocity vectors in the computational plane. Then, Eq. (2) can be expressed in the computational plane as follows

$$\frac{\partial f_\alpha}{\partial t} + \left(\tilde{e}_{\alpha\xi}\frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha\eta}\frac{\partial f_\alpha}{\partial \eta}\right) = -\frac{1}{\tau}\left(f_\alpha - f_\alpha^{eq}\right), \quad \alpha = 0, 1, \ldots, 8 \tag{14}$$

Note that the microscopic velocity vectors $\boldsymbol{e}_\alpha$ are constant on the physical plane, however the transformed microscopic velocity vectors in the computational plane, the contravariant velocity vectors $\tilde{\boldsymbol{e}}_\alpha$, are not constant. Fig. 2 shows the $\boldsymbol{e}_1 = (1, 0)$ vector in both the physical and computational planes.

4. Discretization procedure

In this work, the incompressible form of the LB equation is numerically solved by implementing a high-order compact finite-difference scheme in the generalized curvilinear coordinates. Here, the spatial derivatives in the LB equation in the computational plane are discretized by a fourth-order compact finite-difference scheme and the temporal term is discretized with the fourth-order Runge–Kutta scheme to provide an accurate and efficient incompressible flow solver. The LB equation (14) in the two-dimensional curvilinear coordinates can be written in the following form

$$\frac{\partial f_\alpha}{\partial t} = R(f_\alpha) \quad \text{where } R(f_\alpha) = -\left(\tilde{e}_{\alpha\xi}\frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha\eta}\frac{\partial f_\alpha}{\partial \eta}\right) - \frac{1}{\tau}\left(f_\alpha - f_\alpha^{eq}\right) \tag{15}$$

For an accurate numerical solution of time-dependent problems, it is necessary to discretize the temporal term in the LB equation (15) with a high-order accuracy method. The Runge–Kutta methods have better stability properties and allow for larger time steps compared with the other explicit methods. Here, the solution is advanced in the time t using a four-stage Runge–Kutta scheme. By integrating Eq. (15) with respect to the time t with the explicit fourth-order Runge–Kutta scheme, one can obtain

$$\begin{aligned} &f_\alpha^0 = f_\alpha^t \\ &f_\alpha^k = f_\alpha^0 + \zeta_k \Delta t R^{k-1}(f_\alpha), \quad k = 1, 2, 3, 4 \\ &f_\alpha^{t+\Delta t} = f_\alpha^4 \end{aligned} \tag{16}$$

where the parameters ζ_k ($k = 1$, 2, 3 and 4) are taken as $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$ and 1, respectively. Both steady and unsteady flows are considered in the present study. The fourth-order Runge–Kutta scheme applied is shown to be more suitable for an accurate calculation of unsteady flows (see the numerical results section).

Now, the calculation of the first derivative of the distribution function f_α in the ξ-direction (denoted by $f'_\alpha = \partial f_\alpha/\partial\xi$) is performed with the high-order compact finite-difference scheme as [35,36]

$$\gamma f'_{\alpha,i-1} + f'_{\alpha,i} + \gamma f'_{\alpha,i+1} = \frac{a}{4\Delta\xi}(f_{\alpha,i+1} - f_{\alpha,i-1}) + \frac{b}{4\Delta\xi}(f_{\alpha,i+2} - f_{\alpha,i-2}) \tag{17}$$

where the constants a, b and γ determine the spatial order of accuracy of the scheme. Table 1 gives the values of the constants for the fourth- and sixth-order compact finite-difference scheme. Eq. (17) is applied in the computational domain for the grid points $i = 2, \ldots, Imax - 1$ for each j. To form the tridiagonal system of equations during the i sweep, appropriate boundary conditions are to be used for the terms $f'_{\alpha,1}$ and $f'_{\alpha,Imax}$. These terms at the boundary points can be calculated with appropriate one-sided compact relations. The high-order one-sided boundary condition formulas are expressed as [37]

$$f'_{\alpha,1} + \gamma f'_{\alpha,2} = \frac{1}{\Delta\xi}(af_{\alpha,1} + bf_{\alpha,2} + cf_{\alpha,3} + df_{\alpha,4}) \tag{18}$$

$$f'_{\alpha,Imax} + \gamma f'_{\alpha,Imax-1} = -\frac{1}{\Delta\xi}(af_{\alpha,Imax} + bf_{\alpha,Imax-1} + cf_{\alpha,Imax-2} + df_{\alpha,Imax-3}) \tag{19}$$

Table 1
Coefficients of interior high-order schemes for first derivative discretization.

Scheme	γ	a	b
Fourth-order	$\frac{1}{4}$	$\frac{3}{2}$	0
Sixth-order	$\frac{1}{3}$	$\frac{14}{9}$	$\frac{1}{9}$

Table 2
Coefficients of one-sided high-order schemes for first derivative discretization on the boundary.

Scheme	γ	a	b	c	d
Third-order	2	$-\frac{15}{6}$	$\frac{4}{2}$	$\frac{1}{2}$	0
Fourth-order	3	$-\frac{17}{6}$	$\frac{3}{2}$	$\frac{3}{2}$	$-\frac{1}{6}$

Table 3
High-order filter coefficients at interior points.

Order	a_0	a_1	a_2	a_3	a_4
$F(O(\Delta x^6))$	$\frac{11+10a_f}{16}$	$\frac{15+34a_f}{32}$	$\frac{-3+6a_f}{16}$	$\frac{1-2a_f}{32}$	0
$F(O(\Delta x^8))$	$\frac{93+140a_f}{256}$	$\frac{7+18a_f}{16}$	$\frac{-7+14a_f}{32}$	$\frac{1-2a_f}{16}$	$\frac{-1+2a_f}{128}$

Table 2 gives the values of the constants a, b, c, d and γ for the third- and fourth-order one-sided compact finite-difference formulas. Here, the fourth-order compact formula (17) together with the one-sided third-order relations (18) and (19) are used to compute $\partial f_\alpha/\partial\xi$ implicitly for $i = 1, \ldots, Imax$ for each j. The calculation of the first derivative of the distribution function f_α in the η-direction (denoted by $f'_\alpha = \partial f_\alpha/\partial\eta$) in Eq. (15) is performed for $j = 1, \ldots, Jmax$ for each i in the same manner. The metrics appeared in the formulation $(x_\xi, y_\xi, \ldots)$ are also calculated by the fourth-order compact finite-difference formula implemented.

5. Filtering scheme

The centered high-order compact finite-difference scheme applied to solve the LB equation is nondissipative and it is therefore sensitive to numerical instabilities owing to the growth of high-frequency modes. These difficulties originate usually from grid non-uniformity, boundary conditions and also non-linear flow features [38]. The above difficulties can be circumvented by applying a spatial filtering to the solution of the distribution function f_α. Recently, some studies have been performed to link the lattice Boltzmann method with filtering. Ricot et al. [39] investigated the behavior of several explicit filter stencils in three different LBM algorithms. Brownlee et al. [40] introduced some classes of techniques to improve regularization of LBMs by modifying dissipation. In the present work, a high-order implicit filtering technique [41,42] is applied to damp high-frequency oscillations associated with the central differencing of the spatial derivatives in the LB equation. The implicit filters obtain the filtered variables $\hat{f}_\alpha$ from the unfiltered values f_α by solving the following tridiagonal system of equations

$$a_f \hat{f}_{\alpha,i-1} + \hat{f}_{\alpha,i} + a_f \hat{f}_{\alpha,i+1} = \sum_{n=0}^{N} \frac{a_{n,i}}{2}(f_{\alpha,i+n} + f_{\alpha,i-n}) \tag{20}$$

$2N$ is the order of filter with a $2N+1$ point stencil. The $N+1$ coefficients, a_n, $n = 0, 1, \ldots, N$, are derived in terms of a_f with Taylor and Fourier-series analyses [41–43]. Table 3 presents the filtering coefficients for the tridiagonal sixth- or eighth-order filters. The parameter a_f satisfies the inequality $-0.5 < a_f < 0.5$, in which a higher value of a_f gives a less dissipative filter. The filter is typically chosen to be at least two orders of accuracy higher than the difference method of the LB equation. Hera, for interior points, the sixth-order filter is employed for all the calculations.

Special relations are needed at near boundary points due to the relatively large stencil of the filter. Note that the distribution function f_α at the end points, $i = 1, Imax$ and $j = 1, Jmax$, are not filtered. At other near boundary points, where Eq. (20) cannot be used, as proposed in Ref. [42], one can use higher order one-sided formulas which can retain the tridiagonal form of the filter scheme. In the present simulations, the filter is applied to the distribution function f_α, sequentially in the ξ- and η-coordinate directions, once after each time step of the algorithm. The effects of filtering on the accuracy and performance of the solution of the compact finite-difference LB equation in the generalized curvilinear coordinate are studied (see the numerical results section).

6. Boundary conditions implementation

In this section, the implementation of boundary conditions for both the macroscopic and microscopic variables is described for the compact finite-difference LB scheme in the generalized curvilinear coordinate. All boundary conditions used are either of periodic, Dirichlet or of Neumann type. Boundary conditions of the macroscopic variables are usually known for the fluid flow problems considered. Note that the distribution function f_α in the LB equation is not given directly at the boundaries and a special treatment should be used for determining its value based on the macroscopic variables on each boundary.

The distribution function f_α at a boundary node is defined based on the solution of Eq. (15) implemented on the boundaries. On a wall boundary with no-slip condition or at an inlet boundary, the values of the macroscopic velocity components are known. For calculating the pressure, for example on a wall boundary, one can apply the relation $\partial p/\partial y_n = 0$ which can be applied for moderate and high Reynolds number flows. The grid lines are selected to be orthogonal to each wall boundary ($\eta = \text{const.}$) in the present computations and thus one can use the one-sided fourth-order finite-difference formula for discretizing $\partial p/\partial \eta = 0$ in the computational plane

$$p_1 = \frac{1}{125}(240p_2 - 180p_3 + 80p_4 - 15p_5) \tag{21}$$

to obtain the pressure at the desired boundary. The same procedure can be used for determining the pressure on the other boundaries. The calculation of the pressure at the inlet boundary or the velocity components at the outlet is performed by a similar manner. After updating the macroscopic variables, the equilibrium distribution function f^{eq} is calculated by its formula, Eq. (4), by using the specified/extrapolated macroscopic variables at the desired boundary

$$f^{eq}_{\alpha,1} = fun(p_1, u_1, v_1) \tag{22}$$

Now, the distribution function f_α at the new time step is determined by numerically solving Eq. (15) implemented at the boundaries by the same algorithm employed for the interior points. Note that the spatial derivatives $\frac{\partial f_\alpha}{\partial \xi}$ and $\frac{\partial f_\alpha}{\partial \eta}$ in this equation are known from the compact differencing simultaneously at the interior and boundary points.

7. Numerical results

The accuracy and robustness of the high-order accurate numerical scheme implemented to solve the incompressible form of the LB equation in the generalized curvilinear coordinates are demonstrated for different steady and unsteady flow problems. Four test cases are considered for validating the present computations and demonstrating the accuracy and robustness of the solution algorithm: they are unsteady Couette flow and steady flow in a 2-D cavity with non-uniform grid and steady and unsteady flows over a circular cylinder and the NACA0012 hydrofoil at different flow conditions. Results obtained for these test cases are compared with the available numerical and experimental results. A sensitivity study is also performed to evaluate the effects of grid size and filtering on the accuracy and convergence rate of the solution of the compact finite-difference LBM implemented in the generalized curvilinear coordinates. The characteristic velocity and the filtering parameter are set to be $u_0 = 0.1$ and $a_f = 0.49$, respectively, unless otherwise specified.

7.1. Flow in a 2-D cavity

The square lid-driven cavity flow is a benchmark problem for the assessment of many numerical methods due to its simple geometry, well-defined boundary conditions and complicated flow behaviors. Noslip conditions for the velocity components ($u = v = 0$) are considered for all boundaries except for the upper wall where $u = u_0$. The corners of the cavity are singular points and it is difficult to capture the flow phenomena near the corner regions. Consequently, it is desirable to use non-uniform grids and refine the mesh near these corners. Herein, all the computations based on the compact finite-difference lattice Boltzmann method (CFDLBM) implemented are carried out with $u_0 = 0.1$ and the results are presented for different Reynolds numbers, $100 \leqslant \mathrm{Re} = u_0 L/\upsilon \leqslant 5000$. The non-uniform mesh generated in the wall-normal direction is based on the transformation proposed in [44]:

$$x \quad \text{and} \quad y = L\frac{(2\alpha+\beta)[(\beta+1)/(\beta-1)]^{(\eta-\alpha)/(1-\alpha)} + 2\alpha - \beta}{(2\alpha+1)\{1+[(\beta+1)/(\beta-1)]^{(\eta-\alpha)/(1-\alpha)}\}}, \quad 0 \leqslant \eta \leqslant 1 \tag{23}$$

where β is the clustering parameter, and α defines where the clustering takes place. Herein, the clustering is distributed equally at $x = 0$, $x = L$, $y = 0$ and $y = L$ with $\alpha = 0.5$ and $\beta = 1.08$. Fig. 3 shows a $(101, 101)$ non-uniform mesh in which the grid lines are clustered near the walls.

A grid refinement study is performed for the flow condition $\mathrm{Re} = 100$ to examine the sensitivity of the results to the grid size. The computational grids used for this flow condition is $(Imax, Jmax) = (51, 51), (101, 101), (201, 201)$. Fig. 4 shows both the u-velocity profile along the vertical line and the v-velocity profile along the horizontal line passing through the center of the cavity for the different meshes with $\mathrm{Re} = 100$. The study indicates that the grid (101×101) is appropriate for an accurate solution of the cavity flow by implementing the compact finite-difference lattice Boltzmann method (CFDLBM) for this Reynolds number. The sensitivity study has shown that for $\mathrm{Re} = 100$ and 1000 the grid (101×101) and for $\mathrm{Re} = 5000$ the grid (201×201) are appropriate ones for an accurate calculation of the flowfield using non-uniform grids used.

Fig. 5 gives the computed flowfield shown by the streamlines and the comparison of the u- and v-velocity profiles at the mid-planes for 2-D cavity flow using the present solution procedure for the cavity with $\mathrm{Re} = 100$, 1000 and 5000. The

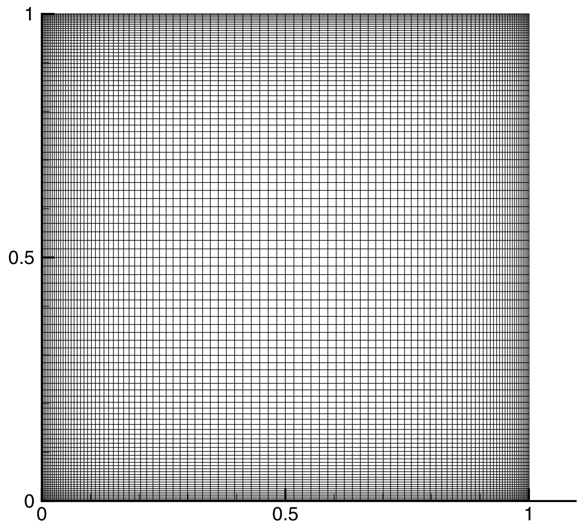

Fig. 3. A (101×101) non-uniform mesh distribution for the lid-driven cavity flow.

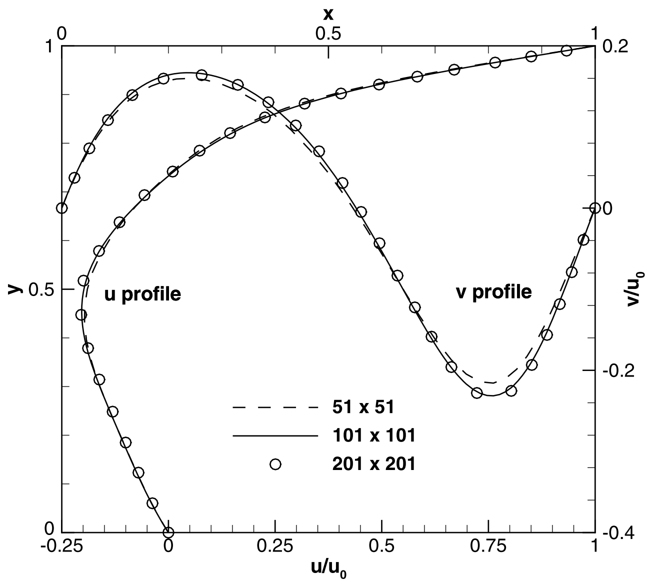

Fig. 4. Grid refinement study on velocity profiles at the mid-planes of 2-D cavity flow with $\mathrm{Re} = 100$.

effect of the value of Reynolds number on the flow structure is observed clearly in this figure. For flows with $\mathrm{Re} = 100$ and 1000, three vortices appear in the cavity: a primary one at the center and a pair of secondary ones in the lower corners of the cavity. As Re increases to 5000, another recirculating region is seen in the upper left corner and a tertiary vortex appears in the lower right corner. It is observed that the flow structure is reasonably resolved by the high-order compact finite-difference LBM implemented on non-uniform grids. The present solution procedure with the non-uniform grid distribution used near the walls allows to calculate the flowfield accurately with a fewer number of grid points. For example, the calculations have shown that for $\mathrm{Re} = 5000$ with uniform grids, a finer grid i.e. (301×301) should be used for an accurate prediction of the flowfield. Both the u-velocity profile along the vertical line and the v-velocity profile along the horizontal line passing through the center of the cavity for each case are shown in Fig. 5. The present results using the fourth-order compact finite-difference LBM are compared with the second-order accurate solution by Ghia et al. [45] and the fourth-order accurate solutions by Erturk et al. [46] and Hejranfar and Khajeh-Saeed [47] which exhibit excellent agreement. It is noted that the solution procedure presented, that is the compact finite-difference LBM, is accurate and robust even at high Reynolds number flows and it provides the results comparable to those of high-order compact finite-difference Navier–Stokes flow solvers.

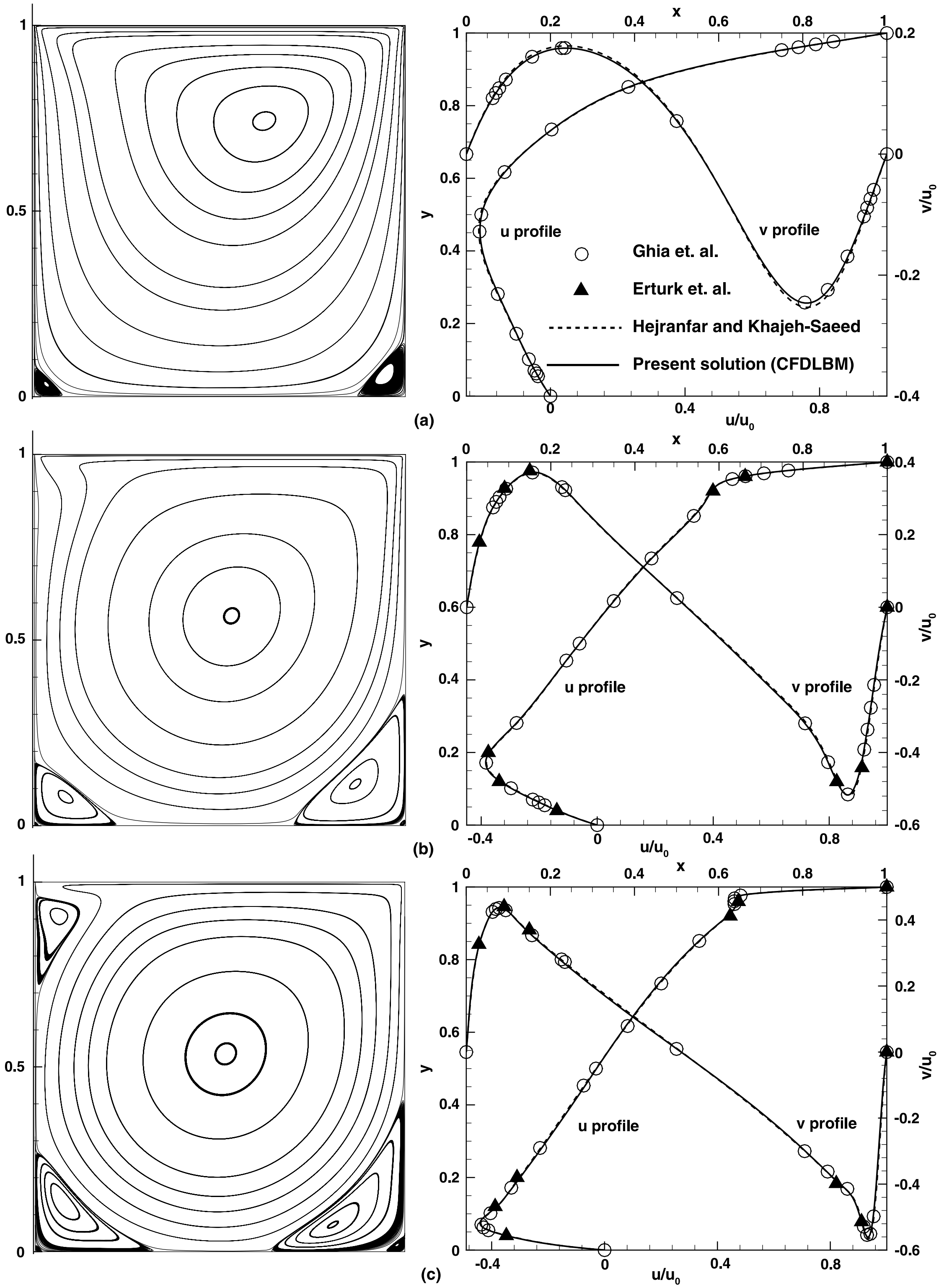

Fig. 5. Computed flowfield and comparison of velocity profiles at the mid-planes for 2-D cavity flow with (a) Re = 100, (b) Re = 1000, (c) Re = 5000.

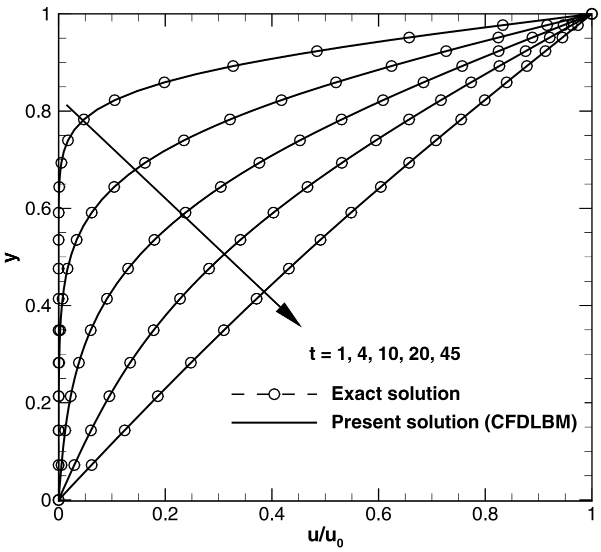

Fig. 6. Comparison of u-velocity profile for the Couette flow with $\mathrm{Re} = 10$.

Table 4
Order of spatial accuracy of the numerical method implemented based on L_2 norm of u-velocity profile for the Couette flow with $\mathrm{Re} = 10$ at $t = 20$.

Grid	Δy	$\mathrm{Log}(\Delta y)$	$\mathrm{Log}(L_2\ \mathrm{norm[Error]})$
(11×21)	1/20	−1.30102	−1.5293
(21×41)	1/40	−1.60205	−2.7147
(41×81)	1/80	−1.90308	−3.8868
Order of spatial accuracy			~ 3.91

7.2. The Couette flow

Now, it is interesting to check the accuracy and efficiency of the compact finite-difference LBM implemented for predicting the time-dependent flow problems. The unsteady incompressible Couette flow between two parallel plates is solved using a non-uniform grid to demonstrate the accuracy and performance of the present solution methodology. A periodic boundary condition is applied to the left and right boundaries and the flow is driven by the upper plate moving along the x-direction with a constant velocity ($u_0 = \mathrm{const.}$, $v_0 = 0$). The present numerical results are compared with the analytical solution for this problem. The exact solution of the normalized streamwise velocity profile for the plane Couette flow is given by the following relation [48]:

$$\frac{u}{u_0} = \frac{y}{H} - \frac{2}{\pi} \sum_{n=1}^{\infty} \exp\left(-n^2 \pi^2 \frac{\mu t}{H^2}\right) \sin\left[n\pi\left(1 - \frac{y}{H}\right)\right] \tag{24}$$

The present calculations are performed for the three computational grids, namely, (11×21), (21×41) and (41×81) at the Reynolds number $\mathrm{Re} = u_0 H/\upsilon = 10$ to obtain the order of accuracy of the solution compared with the analytical solution. The non-uniform mesh in the wall-normal direction is generated by the following transformation:

$$x = \xi$$
$$y = H \frac{(\beta + 1) - (\beta - 1) \times [(\beta + 1)/(\beta - 1)]^{(1-\eta)}}{1 + [(\beta + 1)/(\beta - 1)]^{(1-\eta)}} \tag{25}$$

where the clustering parameter is set to be $\beta = 1.2$. The grid is uniform in the x-direction and it is clustered near the upper wall. Fig. 6 shows a sequence of the normalized streamwise velocity profiles for the grid size (21×41) at different times. It is obvious that the results obtained by applying the compact finite-difference lattice Boltzmann method (CFDLBM) are in good agreement with the exact solution. This simulation is performed with the characteristic velocity $u_0 = 0.001$.

The order of spatial accuracy of the implemented numerical scheme is calculated for this test case based on the calculated streamwise velocity profile compared with the analytical solution. The calculations are performed for the three computational grids at the Reynolds number $\mathrm{Re} = 10$ and $t = 20$. The results in Table 4 show that the order of accuracy of the solution is 3.91 which indicates the fourth-order accuracy of the numerical method implemented for the simulation of

Table 5
Order of temporal accuracy of the numerical method implemented based on L_2 norm of u-velocity profile for the Couette flow with $\text{Re} = 10$ at $t = 20$.

Δt	$\text{Log}(\Delta t)$	1st-order explicit method	4th-order Runge–Kutta method
		$\text{Log}(L_2\ \text{norm[Error]})$	$\text{Log}(L_2\ \text{norm[Error]})$
0.001	−3.00000	–	−2.7147
0.0004	−3.39794	−3.0085	−4.3790
0.0002	−3.69897	−3.3065	−5.5795
0.0001	−4.00000	−3.5676	−6.7649
Order of temporal accuracy		∼0.93	∼3.96

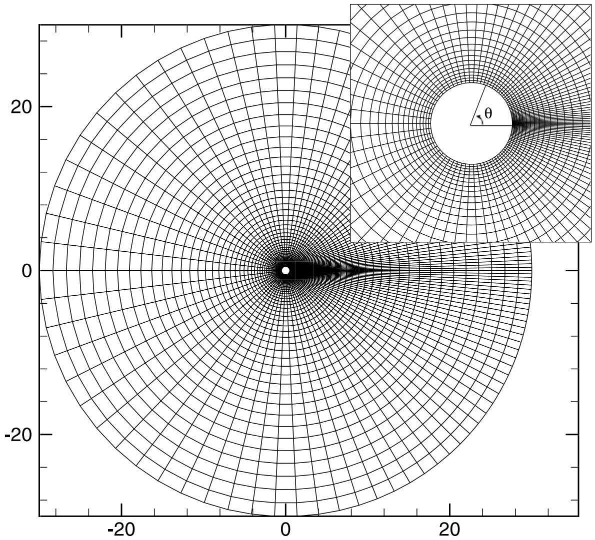

Fig. 7. A (101×51) mesh distribution around the circular cylinder.

time-dependent laminar flows. The error is calculated based on the L_2 norm of the streamwise velocity profile compared with the analytical solution.

To examine the temporal accuracy of the present numerical scheme, numerical experiments are carried out by applying the fourth-order Runge–Kutta scheme compared with the first-order explicit time integration method for the solution of the unsteady Couette flow. The calculations are performed with the computational grid (21×41) for the different time steps Δt for the Reynolds number $\text{Re} = 10$ at $t = 20$. Table 5 verifies the order of accuracy of each time integration scheme and indicates that by using the fourth-order Runge–Kutta scheme, the value of the error is lower than that of the explicit time integration method for each time used. Note also that the fourth-order Runge–Kutta scheme has a larger maximum allowable time step compared with the first-order explicit time integration scheme. This investigation shows that the fourth-order Runge–Kutta scheme used is more appropriate for accurate unsteady simulations.

7.3. Flow over a 2-D circular cylinder

To demonstrate the capability of the present solution procedure for the numerical simulation of the fluid flow around geometries with curved boundaries, the two-dimensional steady and unsteady flows over a circular cylinder is studied at different Reynolds numbers. Here, the Reynolds number is defined as $\text{Re} = u_0 D/\upsilon$ where u_0 is the freestream velocity (here $u_0 = 0.1$) and D is the diameter of the cylinder. This problem has been studied extensively as a benchmark test case by many researchers and thus there are numerous experimental and numerical results available in the literature for this test case for the validation of the results obtained by applying the compact finite difference LBM in the generalized curvilinear coordinates. At low Reynolds numbers, when the Reynolds number is less than its critical value ($\text{Re}_{cr} \approx 47$), a steady state condition is obtained and a pair of stationary recirculating regions appear behind the circular cylinder. The present computations based on the compact finite-difference lattice Boltzmann method (CFDLBM) are performed for the Reynolds numbers $\text{Re} = 10$, 20, 40 and the results obtained are compared with the numerical and experimental results. A sensitivity study is also performed to examine the effects of the grid size and the filtering parameter a_f on the accuracy and the convergence rate of the solution.

The geometry and the near-field detail of the O-grid type with the size of (101×51) used for the numerical simulation of the steady flows around the circular cylinder are shown in Fig. 7. For this test case, the radius of computational domain is set to be 30 times the diameter of the cylinder. The mesh is stretched in the ξ-direction near the rear stagnation point.

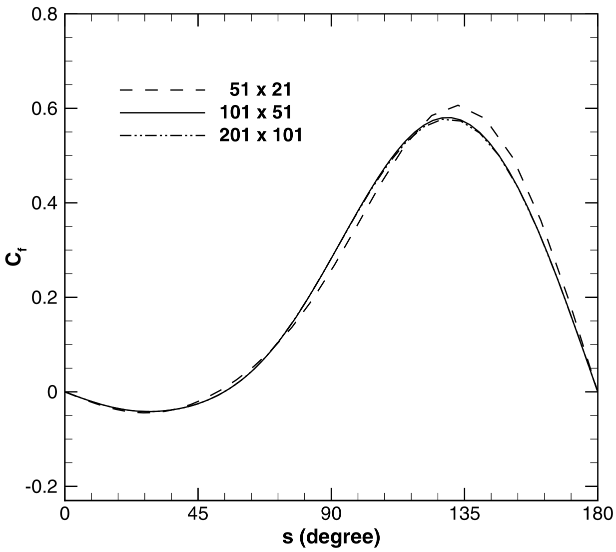

Fig. 8. Grid refinement study on skin friction coefficient distribution for the circular cylinder with Re = 40.

Table 6
Order of spatial accuracy of the numerical method implemented based on error of stagnation pressure coefficient at the front of the cylinder, $C_p(\pi)$, with Re = 40.

Grid	Δs	$\mathrm{Log}(\Delta s)$	$\mathrm{Log}(\mathrm{Error}[C_p(\pi)])$
(51×21)	$2\pi/50$	-0.90079	-1.8246
(101×51)	$2\pi/100$	-1.20182	-2.8119
(201×101)	$2\pi/200$	-1.50285	-4.0991
Order of spatial accuracy			$\sim$3.77

For the stretching in the η-direction which is in the wall-normal direction here, the following transformation function is used [44]:

$$y_n = \delta \frac{(\beta + 1) - (\beta - 1) \times [(\beta + 1)/(\beta - 1)]^{(1-\eta)}}{1 + [(\beta + 1)/(\beta - 1)]^{(1-\eta)}}, \quad 0 \leqslant \eta \leqslant 1 \tag{26}$$

where δ is the radial distance between the body and outer boundary. The stretching parameter β is set to be 1.06 for all the cases computed.

At first, three different computational meshes, the mesh I (51×21), mesh II (101×51) and mesh III (201×101), are used to examine the sensitivity of the results on the grid size and also verify the order of accuracy of the solution. Fig. 8 gives a grid refinement study on the skin friction coefficient distribution with Re = 40 calculated by the fourth-order compact finite-difference LBM. The study demonstrates that the mesh II is appropriate for an accurate computation of the flowfield and a further grid refinement does not significantly improve the accuracy of the results. Thus, the mesh II is considered as a suitable computational grid, regarding the accuracy and the computational effort, for all the steady flow calculations around the circular cylinder. In Table 6, the order of spatial accuracy of the solution for this test case is calculated for Re = 40 based on the L_2 norm of the surface pressure coefficient distribution for all the grid points for the meshes (51×21), (101×51) and (201×101), compared with the results of the most refined one, i.e., the mesh (301×201). The order of spatial accuracy is about 3.77 that verifies the fourth-order accuracy of the numerical scheme implemented.

To check the necessity of using the low-pass filter, the simulation of flowfield for the circular cylinder test case is investigated for different Reynolds numbers without filtering. Numerical experiments have indicated that the instabilities are observed without using the low-pass filter that causes the solution to be unstable after a few iterations. A sensitivity study is conducted to investigate the effects of the value of the filtering coefficient α_f on the accuracy and convergence rate of the solution with Re = 40. This analysis is also conducted to evaluate the upper limit of the value of the filtering coefficient α_f (close to $\alpha_f = 0.5$) at which the scheme becomes unstable. Fig. 9 shows the skin friction coefficient distribution over the circular cylinder at Re = 40 for different values of the filtering coefficient α_f. As shown in this figure, the accuracy of the solution computed based on the compact finite-difference LBM applied is not affected by the value of the filtering parameter for $\alpha_f \geqslant 0.49$. For this case, by selecting the filtering parameter $\alpha_f \leqslant 0.49999$ (very close to $\alpha_f = 0.5$) the scheme is still stable and it becomes unstable for $\alpha_f \geqslant 0.49999$. The filtering parameter is set to be $\alpha_f = 0.49$ in all the problems studied in the paper to ensure the solution to be stable in the all conditions. Fig. 10 illustrates the convergence history of the solution for both the L_2 norm and L_∞ norm (maximum norm) based on the u-velocity profile in the flowfield at

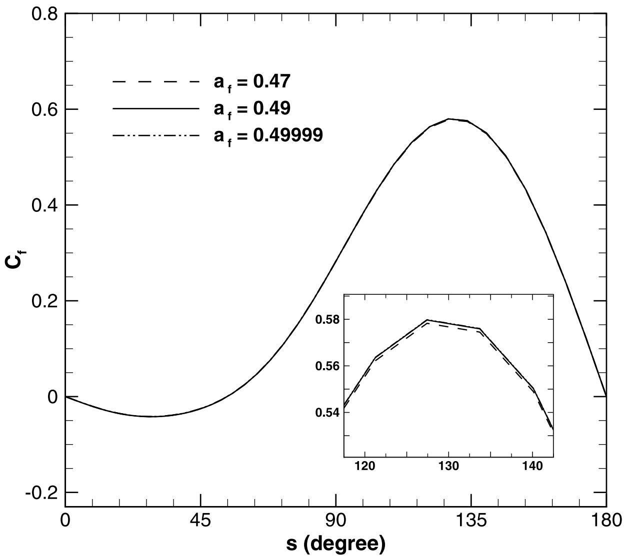

Fig. 9. Effect of value of filtering coefficient a_f on skin friction coefficient distribution for the circular cylinder with $\mathrm{Re} = 40$.

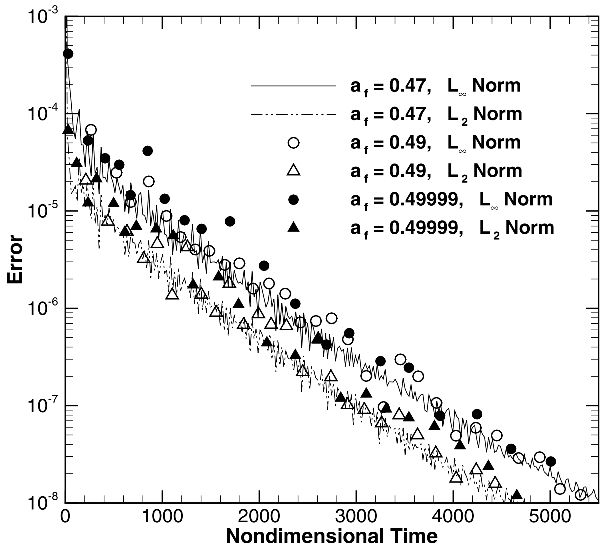

Fig. 10. Effect of value of filtering coefficient a_f on convergence history of the solution for the circular cylinder with $\mathrm{Re} = 40$.

$\mathrm{Re} = 40$. The study indicates that the convergence rate to the steady-state solution is not also very sensitive to the value of the filtering parameter for $a_f \geqslant 0.49$.

The results for the steady flow past the circular cylinder are presented for different Reynolds numbers. Fig. 11 shows the computed flowfield depicted by the streamlines for $\mathrm{Re} = 10$, 20 and 40. It is observed that a pair of symmetric stationary recirculating regions appears in the wake of the cylinder at the three Reynolds numbers considered. As the Reynolds number increases, the length of the recirculating region, L, (the distance from the rearmost point of the cylinder to the end of the wake) and the separation angle, θ_s, increase. Table 7 compares the computed results by applying the compact finite-difference LBM in the curvilinear coordinates with the available experimental and numerical results [1,10,12,49–53] for the hydrodynamic coefficients and the locations of the separation and reattachment points for $\mathrm{Re} = 10$, 20 and 40. Dennis and Chang [51] and Nieuwstadt and Keller [52] used a finite-difference method to solve the Navier–Stokes equations for predicting laminar flow over the circular cylinder. The results by Mei and Shyy [10] and Guo and Zhao [12] were obtained by finite-difference LBM (FDLBM). Also, the results by He et al. [53] and Imamura et al. [1] were obtained by the interpolation supplemented LBM (ISLBM) and the generalized form of ISLBM (GILBM), respectively. As shown in Table 7, the present results obtained by applying the compact finite-difference LBM for the drag coefficient show similar trends relative to the values obtained by other methods. The stagnation pressure coefficient at the front of the cylinder, $C_p(\pi)$, and at its rear, $C_p(0)$, are also compared well with the results of the previous studies. The values of the length of the recirculating region

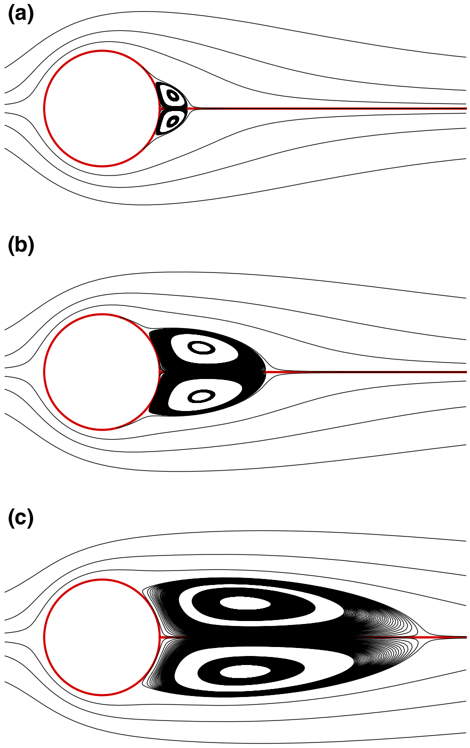

Fig. 11. Computed flowfield for steady flow over the circular cylinder shown by streamlines with (a) Re = 10, (b) Re = 20, and (c) Re = 40.

Table 7
Effect of value of Reynolds number on the results of steady flow over the circular cylinder and comparisons with available results.

Re	Author(s)	Method	C_d	$-C_p(0)$	$C_p(\pi)$	$2\frac{L}{D}$	θ_s
10	Coutanceau and Bouard	Experiment	–	–	–	0.68	32.5
	Tritton	Experiment	3.06	–	–	–	–
	Dennis and Chang	N-S	2.846	0.742	1.489	0.53	29.6
	Nieuwstadt and Keller	N-S	2.828	0.692	1.500	0.434	27.96
	Guo and Zhao	FDLBM	3.049	0.661	1.476	0.486	28.13
	Mei and Shyy	FDLBM	–	–	–	0.489	30.0
	He et al.	ISLBM	3.170	0.687	1.393	0.474	26.89
	Imamura et al.	GILBM	2.848	0.733	1.403	0.478	26.0
	Present solution	CFDLBM	2.801	0.6733	1.4867	0.4891	29.83
20	Coutanceau and Bouard	Experiment	–	–	–	1.86	44.8
	Tritton	Experiment	2.2	–	–	–	–
	Dennis and Chang	N-S	2.045	0.589	1.269	1.88	43.7
	Nieuwstadt and Keller	N-S	2.053	0.582	1.274	1.786	43.37
	Guo and Zhao	FDLBM	2.048	0.512	1.289	1.824	43.59
	Mei and Shyy	FDLBM	–	–	–	1.804	42.1
	He et al.	ISLBM	2.152	0.567	1.233	1.842	42.96
	Imamura et al.	GILBM	2.051	0.589	1.251	1.852	43.3
	Present solution	CFDLBM	2.021	0.5465	1.2659	1.8480	43.58
40	Coutanceau and Bouard	Experiment	–	–	–	4.26	53.5
	Tritton	Experiment	1.65	–	–	–	–
	Dennis and Chang	N-S	1.522	0.509	1.144	4.69	53.8
	Nieuwstadt and Keller	N-S	1.550	0.554	1.117	4.357	53.34
	Guo and Zhao	FDLBM	1.475	0.448	1.168	4.168	52.44
	Mei and Shyy	FDLBM	–	–	–	4.38	50.12
	He et al.	ISLBM	1.499	0.487	1.113	4.490	52.84
	Imamura et al.	GILBM	1.538	0.514	1.156	4.454	52.4
	Present solution	CFDLBM	1.515	0.4808	1.154	4.510	51.86

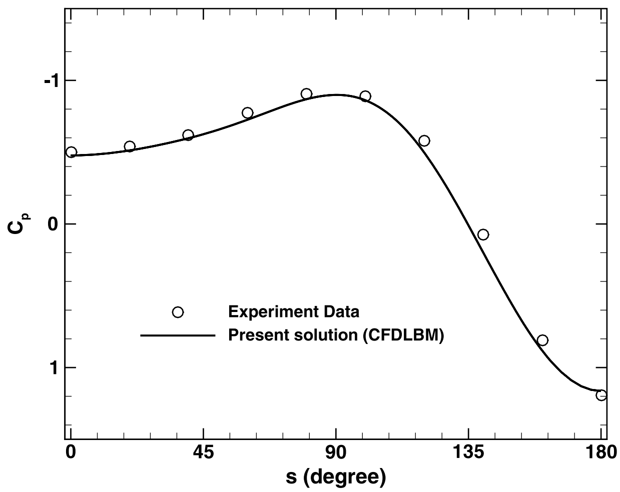

Fig. 12. Comparison of surface pressure coefficient distribution for the circular cylinder with $\mathrm{Re} = 40$.

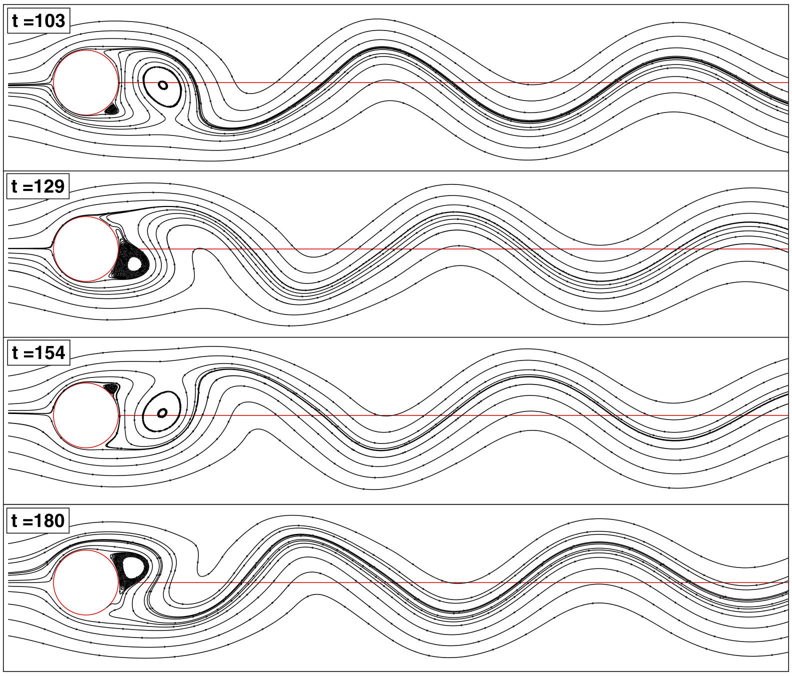

Fig. 13. Computed flowfield for unsteady periodic flow over the circular cylinder shown by streamlines with $\mathrm{Re} = 200$.

normalized by the radius of the cylinder, $2L/D$, and the separation angle are comparable with those of finite-difference Navier–Stokes solvers and finite-difference LBM for the three Reynolds numbers studied. Fig. 12 shows the surface pressure coefficient distribution obtained by employing the compact finite-difference LBM in comparison with the experimental data [54] for the Reynolds number 40, which exhibits good agreement.

To demonstrate the capability of the compact finite-difference LBM implemented to simulate time-dependent flows around the geometries with curved boundaries, unsteady laminar flows over a circular cylinder with moderate Reynolds numbers are also simulated. The cylinder wake becomes unstable for $\mathrm{Re} \geqslant 47$. At this condition, the stationary vortices behind the cylinder become unsteady and a periodic vortex shedding is observed. The Strouhal number is defined as the dimensionless vortex shedding frequency $St = f_q D/u_0$ based on the shedding frequency f_q. By the definition of the dimensionless time of shedding period, $T_p = u_0 t/D$, the Strouhal number can be expressed as $St = D/u_0 T_p$. Herein, D is the diameter of the cylinder and T_p is the peak-to-peak period of the lift or drag coefficient. This study is performed using the grid size of (201×101) and the radius of the computational domain is 30 times the diameter of the cylinder.

Two Reynolds numbers, namely 100 and 200, are chosen to compare the present unsteady results with the existing experimental and numerical ones. Comparisons are carried out for the lift and drag coefficients and also for the value

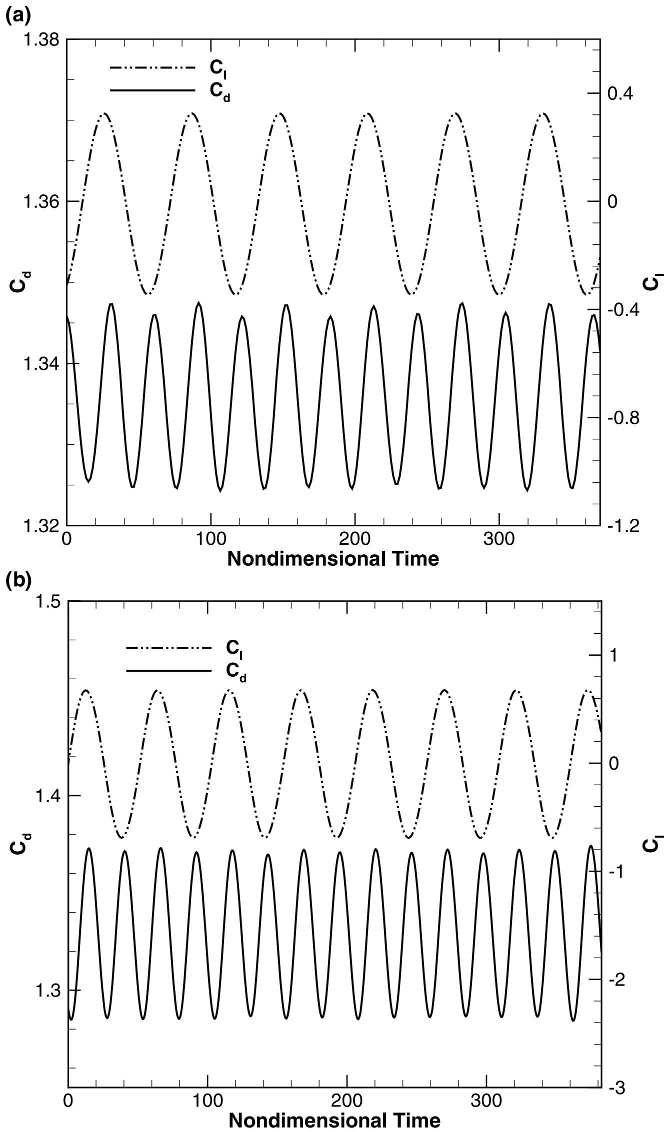

Fig. 14. Time variation of drag and lift coefficients for unsteady periodic flow over the circular cylinder with (a) Re = 100 and (b) Re = 200.

of Strouhal number. Fig. 13 shows the instantaneous streamlines for the developed periodical flow with Re = 200 behind the cylinder in one cycle of vortex shedding, after the dynamically periodic solution is established. The predicted vortex shedding demonstrates that the compact finite-difference LBM implemented well-predicts the unsteady flow patterns for the circular cylinder. The time evolution of the drag and lift coefficients are shown in Fig. 14 for Re = 100 and 200. In this figure, the periodicity can be observed clearly. Table 8 presents a comparison of the calculated drag and lift coefficients and also the Strouhal number using the high-order compact finite-difference LBM with those of the experimental [55] and numerical [56–60] results at Re = 100 and 200. The range of the results predicted by the present simulations is in good agreement with the available results.

7.4. Flow around the NACA0012 hydrofoil

The numerical simulation of the steady and unsteady laminar flows around the NACA0012 hydrofoil is performed to demonstrate the accuracy and robustness of the solution methodology based on the compact finite-difference LBM implemented in the generalized curvilinear coordinates. The Reynolds number, $\mathrm{Re} = u_0 c/\upsilon$, is defined based on the freestream velocity u_0 and the chord length of the hydrofoil c. The Poisson grid generation method based on the work of Steger and Sorenson [61] is employed to generate the grids for this test case. The grid lines are nearly orthogonal to the body surface with an appropriate clustering for accurately resolving the flowfield near the body surface. The geometry of the NACA0012 hydrofoil and the near-field detail of the O-grid type with the size of (101×71) used for the simulation of the steady flows

Table 8
Comparison of the predicted hydrodynamic coefficients and Strouhal number for unsteady flow over the circular cylinder with Re = 100 and 200.

Re	Author(s)	Method	C_d	C_l	St
100	Liu et al.	N-S	1.35 ± 0.012	± 0.339	0.164
	Calhoun	N-S	1.35 ± 0.014	± 0.3	0.175
	Russell and Wang	N-S	1.38 ± 0.007	± 0.322	0.169
	Chiu et al.	N-S	1.35 ± 0.012	± 0.303	0.167
	Present solution	CFDLBM	1.3360 ± 0.0114	± 0.3354	0.1626
200	Berger and Wille	Experiment	1.30	–	0.19
	Liu et al.	N-S	1.31 ± 0.049	± 0.69	0.192
	Wright and Smith	N-S	1.33 ± 0.04	± 0.68	0.196
	Calhoun	N-S	1.17 ± 0.058	± 0.67	0.202
	Russell and Wang	N-S	1.29 ± 0.022	± 0.5	0.195
	Chiu et al.	N-S	1.37 ± 0.051	± 0.71	0.198
	Present solution	CFDLBM	1.329 ± 0.0439	± 0.6814	0.1941

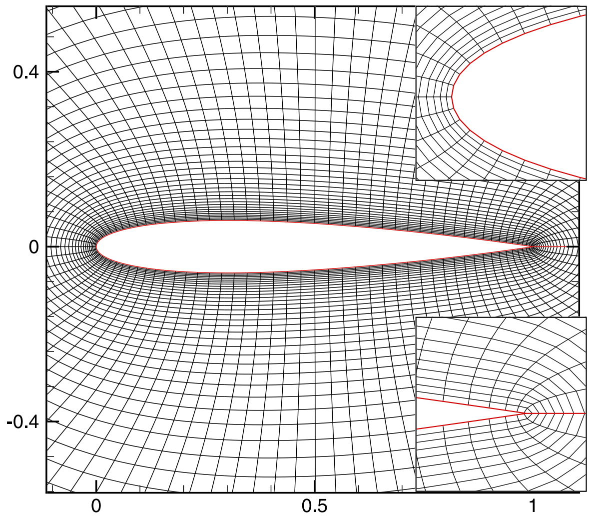

Fig. 15. Computational O-grid used in simulation of steady laminar flow over the NACA0012 hydrofoil with a (101×71) resolution.

are shown in Fig. 15. For this geometry, the radius of computational domain is set to be 20 chords. This study is performed with Re = 500 and at two angle-of-attacks $\alpha = 0^\circ$ and 10°.

A grid refinement study is performed to assess the grid resolution requirements around the NACA0012 hydrofoil at 10 degrees angle-of-attack. Three grid systems, the coarse grid (61×41), the medium grid (101×71) and the fine grid (201×121); are used for this study. As illustrated in Fig. 16, the differences between the predicted surface pressure and skin friction coefficient distributions for the medium and fine meshes are small and the solution seems independent of the grid size when the number of grid points become larger than the medium one. Thus, the medium mesh system is utilized for the all subsequent calculations presented here because it has less computational effort without affecting the accuracy of the solution.

The present results based on implementing the compact finite-difference LBM in the curvilinear coordinates are compared with the results of Hafez et al. [62] in Fig. 16. They used a viscous/inviscid interaction procedure coupled with the potential flow and boundary layer calculations to numerically simulate steady and unsteady laminar flows over the NACA0012 hydrofoil. The study shows that the implemented algorithm provides accurate and reliable results for this geometry at relatively high angle-of-attacks.

Fig. 17 shows the pressure contours and the streamlines obtained by the present solution procedure for the flow around the NACA0012 hydrofoil with Re = 500 at the angle-of-attacks $\alpha = 0^\circ$ and 10°. A symmetry flow pattern is observed at $\alpha = 0^\circ$, while the flow separation is occurred in the suction side of the hydrofoil at $\alpha = 10^\circ$. In Fig. 18, the surface pressure coefficient distribution and the velocity profiles at $x/c = 0.5$ calculated based on the present compact finite-difference LBM are compared with the results of generalized form of interpolation supplemented LBM (GILBM) and the CFL3D code obtained by Imamura et al. [63] at Re = 500 and $\alpha = 0^\circ$. The CFL3D code is a Navier–Stokes flow solver based on the finite-volume formulation. As shown in this figure, the results obtained agree well with those of GILBM and CFL3D which indicates the validity of the simulations performed by the compact finite-difference LBM.

To more assess the accuracy and performance of the present compact finite difference LBM implemented in the curvilinear coordinates, the numerical solution of unsteady laminar flow over the NACA0012 hydrofoil is also performed. The

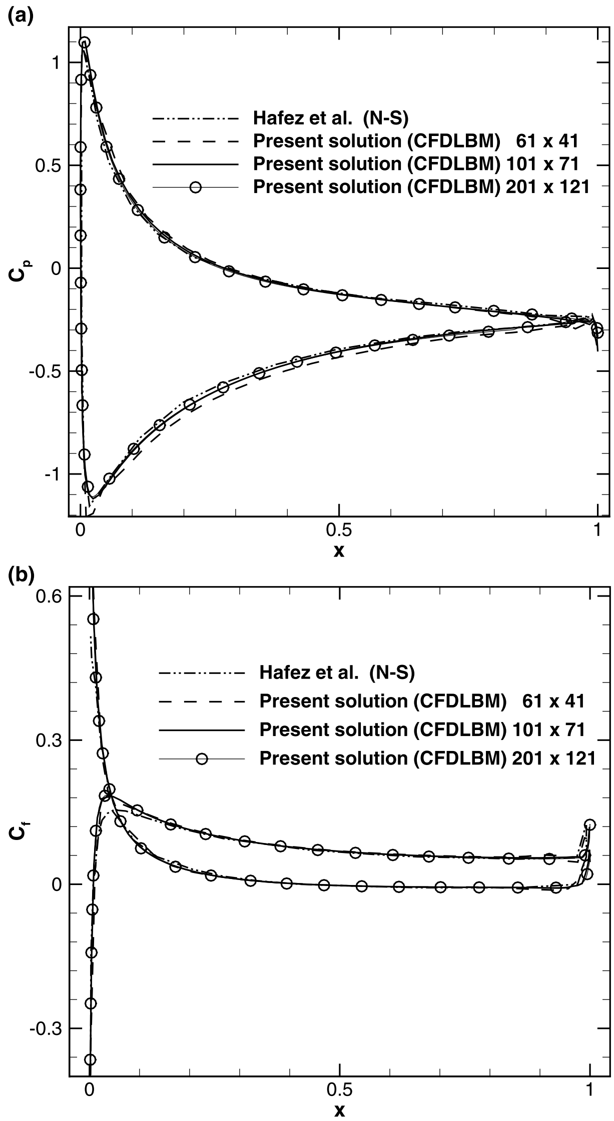

Fig. 16. Grid refinement study on (a) surface pressure coefficient and (b) skin friction coefficient distributions for the NACA0012 hydrofoil with $\mathrm{Re} = 500$ and $\alpha = 10°$.

laminar flow over the NACA0012 hydrofoil at the Reynolds number $\mathrm{Re} = 800$ and the angle-of-attack $\alpha = 20°$ is also simulated by Hafez et al. [62]. At this condition, the flow over the hydrofoil becomes periodically unsteady with the development of a vortex street. Here, the computations are performed with the grid size of (251×121) and the radius of computational domain is set to be 100 chords.

Fig. 19 shows the instantaneous streamlines around the hydrofoil at three different dimensionless times. The low frequency periodical vortex shedding is clearly observed in this figure. The temporal development of the v-velocity component at the point $(x/c, y/c) = (1.1, 0.0)$ behind the NACA0012 hydrofoil calculated based on the present solution procedure are given in Fig. 20 and compared with that of Hafez et al. The results obtained by the present solution and that of Hafez et al. are almost identical and the small difference observed between these two solution procedures may be due to different numerical algorithms with different accuracies or different grid sizes/distribution used.

8. Conclusion

In this work, a high-order compact finite-difference lattice Boltzmann method (CFDLBM) is implemented in the generalized curvilinear coordinates to improve the computational efficiency of the solution algorithm to handle curved geometries with non-uniform grids. The incompressible form of the LB equation with the Bhatnagar–Gross–Krook (BGK) approximation is transformed into the generalized curvilinear coordinates and then the spatial derivatives in the resulting equation in the computational plane are discretized by using the fourth-order compact finite-difference scheme and the temporal term is

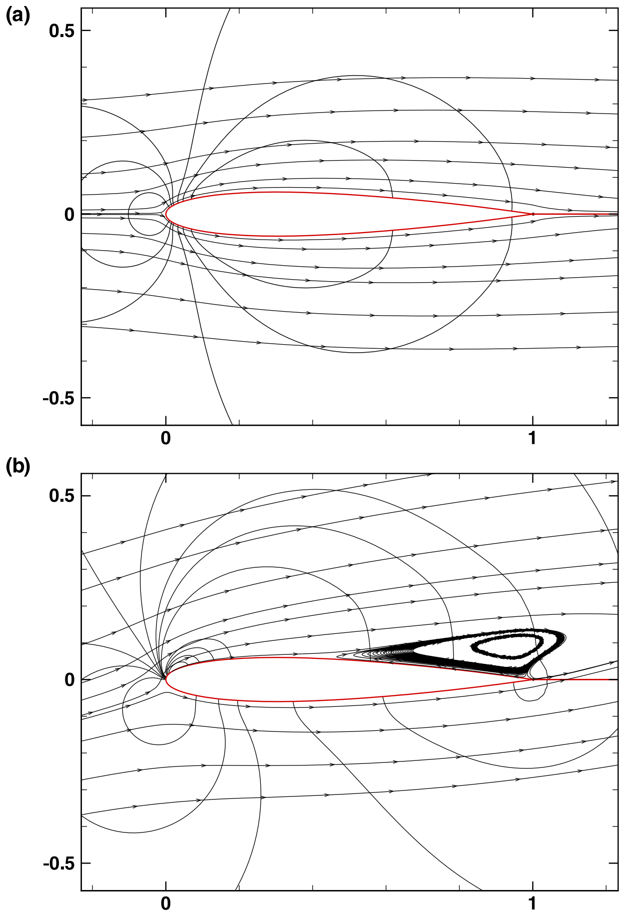

Fig. 17. Computed flowfield for steady flow around the NACA0012 hydrofoil shown by pressure contours and streamlines with $\mathrm{Re} = 500$ at (a) $\alpha = 0°$, and (b) $\alpha = 10°$.

discretized with the fourth-order Runge–Kutta scheme. A high-order spectral-type low-pass compact filter is used to ensure the numerical stability. The calculations are performed for different steady and unsteady incompressible flow benchmark problems to demonstrate the accuracy and robustness of the high-order compact LBM applied.

It is shown that the computed results obtained by implementing the compact finite-difference LBM in the generalized curvilinear coordinates are in good agreement with the analytical solutions and also the available numerical and experimental results for the test cases considered. The sensitivity study indicates that the accuracy and the convergence rate of the solution of the compact finite-difference LBM applied are not very sensitive to the value of filtering parameter. The boundary conditions applied by solving the governing equations in the generalized curvilinear coordinates on the boundaries based on given/calculated macroscopic values of the flow variables can resolve the flowfield near the boundaries accurately. The study shows the present solution methodology based on the implementation of the high-order compact finite-difference Lattice Boltzmann method (CFDLBM) in the generalized curvilinear coordinates is able to accurately and efficiently solve steady and unsteady incompressible flows over practical geometries.

Results obtained by applying the high-order compact finite-difference LBM method in the generalized curvilinear coordinates are also comparable with those of Navier–Stokes solvers. Note that the formulation of the compact finite-difference LBM is simpler to implement and it can be considered as an appropriate alternative computational technique to high-order compact finite-difference Navier–Stokes solvers for precisely studying physical phenomena and simulating fluid dynamics over practical geometries. The extension of the proposed solution procedure to simulate 3-D low speed flows over practical geometries is straightforward and is undergoing.

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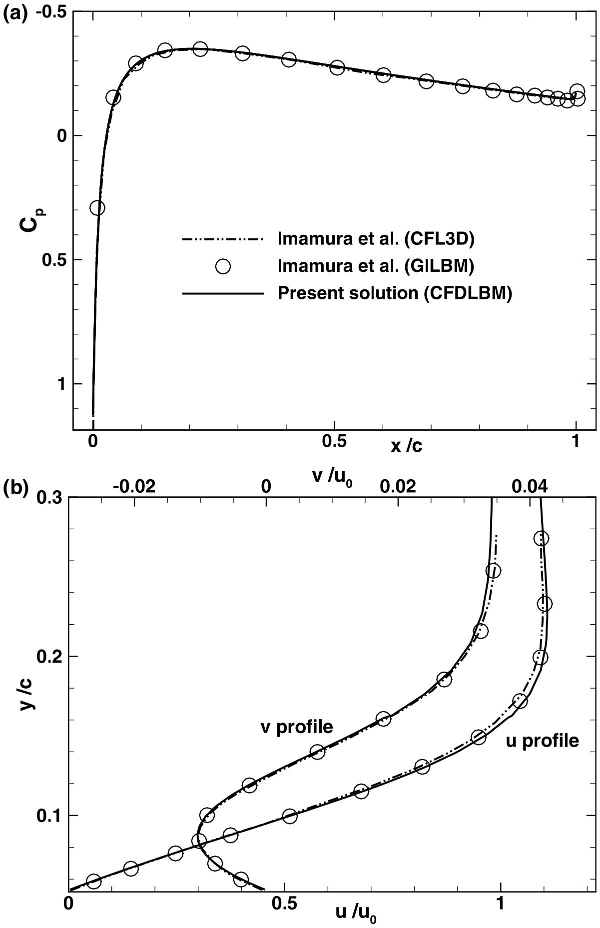

Fig. 18. Comparison of (a) surface pressure coefficient distribution and (b) boundary layer profile at $x/c = 0.5$, for the NACA0012 hydrofoil with $\mathrm{Re} = 500$ at $\alpha = 0°$.

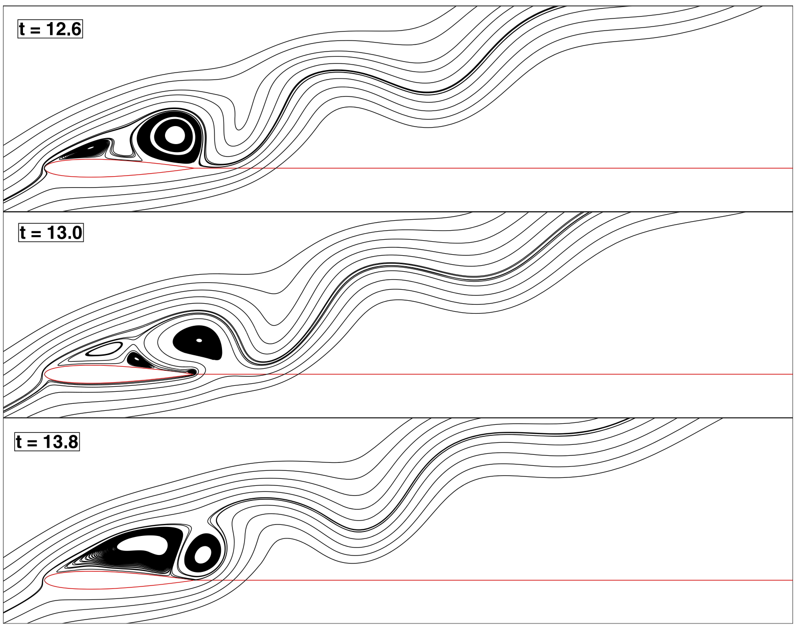

Fig. 19. Computed flowfield for unsteady periodic flow over the NACA0012 hydrofoil shown by streamlines with $\mathrm{Re} = 800$ and $\alpha = 20°$.

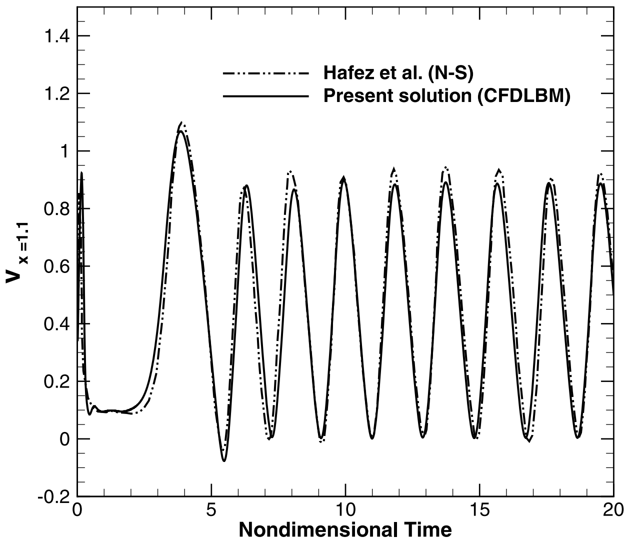

Fig. 20. Comparison of temporal development of v-velocity component at $(x/c, y/c) = (1.1, 0.0)$ for the NACA0012 hydrofoil with $\mathrm{Re} = 800$ and $\alpha = 20°$.

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